

Mark Scheme - Coding Part1 (GANs)

Question	Answer - Total = 45 points
Q1.1: Dataset	Total - 5 points Successful heart shaped data 0 if you decide to use the provided snake dataset
Q1.2: Generate dataset	Total - 5 points Successfully generated data on scatter plot - 3 points Successfully prepared dataloader - 2 points
Q2.1: Generator Net	Total - 5 points Correct in and out features - 1 points Correct hidden0 - 1 points Correct hidden1 - 1 points Correct out - 1 points Correct forward - 1 points
Q2.2: Discriminator Net	Total - 5 points Correct in and out features - 1 points Correct hidden0 - 1 points Correct hidden1 - 1 points Correct out - 1 points Correct forward - 1 points
Create an instance of generator and discriminator	Total - 1 points Correctly created generator instance Correctly created discriminator instance
Q2.3	Total - 2 points Successfully initialized optimizer - 1 point

	Successfully initialized loss function - 1 points
Q2.4.1: Prepare your target	Total - 2 points Correct fake data target generation function
Q2.4.2 Train discriminator	Total - 15 points Propagate real data - 10 points Propagate fake data - 5 points
Successfully get training working	Total - 5 points Successfully get training working

Mark Scheme - Coding Part1 (Diffusion)

Question	Answer - Total = 55 points
Q1.1: Alpha, Alpha bar, q function	Total - 10 points Correctly define alpha - 2 points Correctly define alpha bar- 2 points Correctly define sqrt_ab - 3 points Correctly define sqrt_one_minus_ab- 3 points
Q1.2: Visualize noising	Total - 5 points Correctly define xt
Q2.1: Backwardprocess	Total - 10 points Correctly used time_embed_dim - 2 points Correctly define linear_relu_linear - 5 points Correct forward - 3 points

Q2.2: Discriminator Net	Total - 5 points Correct in and out features - 1 points Correct hidden0 - 1 points Correct hidden1 - 1 points Correct out - 1 points Correct forward - 1 points
Q2.3: sqrt_one_minus_ab, mean	Total - 4 points Correctly defined sqrt_one_minus_ab - 2 points Correctly defined mean - 2 points
Q2.4: sample_ddpm	Total - 1 points Correctly defined sample_ddpm
Q3: Discussion	Total - 20 points - Reflecting on your experience accordingly - 10 points - Suppose you scale up to a much more complex, high-dimensional dataset, reflect and discuss - Would you choose a GAN or a Diffusion Model? - 5 points - Which trade-offs (speed, stability, quality) would matter most in your choice, and why? - 5 points

Mark Scheme - Coding Part2 (ViT)

Question	Answer - Total = 30 points
Change head to out_features 100	Total - 10 points Successfully change head
Load weight	Total - 5 points Successfully load weight

Prepare input	Total - 10 points Perform transformation Convert to tensor
Final prediction	Total - 5 points Successfully get final prediction

Mark Scheme - Coding Part2 (DINO)

Question	Answer - Total = 20 points
Center and sharpen teacher's output	Total - 5 points Successfully done
Sharpen student's outputs	Total - 5 points Successfully done
Calculate loss	Total - 5 points Successfully done
Completeness	Total - 5 points Successfully done